TECHNICAL SUMMARY / ENGINEERING PROTOTYPE

Reef Rescue

Perception-guided intervention, with an evidence trail.

USER & PROBLEM

Reef managers, research groups and restoration teams need repeatable surveys and reviewable interventions despite limits on diver time and access. Reef Rescue explores supervised collection and placement of **already detached, eligible coral fragments**, while recording what the robot saw and why it acted.

CURRENT DEMO ARCHITECTURE

ROBOT CAMERA
Actual rendered RGB
Gemma appearance evidence

→

GATEWAY POLICY
Registry eligibility + cuOpt
Bounded route proposal

→

LOCAL EXECUTION
Freshness and safety gates
Pickup, place, verify

Argon assistance: uncertainty or issues trigger bounded rescan, hold or abort advice; no safety override. Terminal review remains optional.
Gemma: intended onboard; this demo uses Modal. Offline onboard operation is not demonstrated.

MODEL PLACEMENT & MEASURED TIMING

Component	Current execution / hardware	Evidence and latency
Gemma 4 E4B	Remote Modal / vLLM; L4 or A10 configured	2.687-3.201 s, n=5 [1] Browser request round trips
NVIDIA cuOpt	Remote Modal; T4 configured	0.198 s, n=1 [1] Native solve time
Gemini 4 Argon	Google cloud; provider hardware unspecified	22.700 s, n=1 [1] Assistance round trips
Policy / safety	Policy: CPU gateway; safety: browser host	Control latency and power: not yet measured

GPU names are configuration, not measured allocated-device identity. The 1 Hz model-loop target remains unverified; rendering rate is not inference rate.

FEATURES & DECISION EVIDENCE

Sequence: UV-marker inspection, white-light RGB evidence, loose pickup, safe transit, attachment and mechanical verification. Synthetic UV markers do not identify genotype. Camera hashes, capture IDs and telemetry connect evidence to actions; cuOpt compares bounded corridors.

WHAT HAS BEEN DEMONSTRATED

Latest saved run [1]: 1 verified placement(s), 0 drops, 0 recoveries. Final phase: complete. A model response alone does not establish successful handling. **Regression fixtures [2]:** 24 loose-stock pickups across three seeds; current-loss recovery tests use model doubles.

CRITICAL FAILURE: CURRENT-INDUCED LOSS

A surge can exceed modeled grip load. The controller detects loss, tracks the same fragment, requests a fresh view and recovers or aborts. Argon can advise rescan, hold or abort; it cannot override uncertain evidence. Fresh perception, bounded retries, eligibility and local clearance remain mandatory.

LIMITS & NEXT VALIDATION

Reduced browser physics and synthetic sonar are separate from the read-only Isaac RTX feed. RGB does not prove viability or genotype; registered facts are separate. Edge residency, power, calibrated accuracy and field reliability remain unverified. Mechanical placement is not biological recovery.

Field intent: no healthy wild-colony harvesting and no dead-coral revival. Restoration experts must select suitable stock/sites and permissions; outcomes require repeated survival and condition monitoring.

Evidence: 2026-11-08 13:41:14. [1] docs/evidence/live-model-episode.json. [2] sim/pelagic/tests/lab-simulation.test.mjs. Model roles: docs/ARCHITECTURE.md. Sources are repository records, not field-performance estimates.